

Lecture 10: Local Projections

Ko Sugiura

April 30, 2026

Ghent University

Local Projections

Introduction

Recap: Finite-Sample Property of VAR-Based Forecast

- Last week, we studied the VAR-based forecast:

$$IRF_t(h) = B_1^h A_0^{-1}.$$

- In a finite sample, the OLS estimators are not unbiased:

i.e., $\hat{B}_1 = B_1 + \tau_B$, where $E[\tau_B] \neq 0$.

- The impulse response function is biased:

$$\begin{aligned}\widehat{IRF}_t(h) &= \hat{B}_1^h \hat{A}_0^{-1} \\ &= (B_1 + \tau_B)^h \hat{A}_0^{-1} \\ &= \underbrace{(B_1 + \tau_B) \dots (B_1 + \tau_B)}_{h \text{ biases}} \hat{A}_0^{-1}\end{aligned}$$

Outline

- Preliminary: Statistics
- Definition
- Identification
- Estimation and Inference

Preliminary

Statistics

Preliminary: Bias-Variance Tradeoff

Fact: Bias-Variance Tradeoff

Let $\hat{y}_{t+1|t}$ be a (one-step-ahead) forecast for y_{t+1} . Then, the mean-squared forecast error of $\hat{y}_{t+1|t}$, $MSFE(\hat{y}_{t+1|t})$, can be decomposed as

$$MSFE(\hat{y}_{t+1|t}) = Var(y_{t+1}) + Var(\hat{y}_{t+1|t}) + (Bias(\hat{y}_{t+1|t}))^2.$$

Sketch of proof. Conditional on $\mathcal{I}_t$,

$$\begin{aligned} \underbrace{E[(y_{t+1} - \hat{y}_{t+1|t})^2]}_{MSFE(\hat{y}_{t+1|t})} &= \underbrace{E[y_{t+1}^2]}_{Var(y_{t+1}) + E[y_{t+1}]^2} - 2 \underbrace{E[y_{t+1}\hat{y}_{t+1|t}]}_{Cov(y_{t+1}, \hat{y}_{t+1|t}) + E[y_{t+1}]E[\hat{y}_{t+1|t}]} + \underbrace{E[\hat{y}_{t+1|t}^2]}_{Var(\hat{y}_{t+1|t}) + E[\hat{y}_{t+1|t}]^2} \\ &= Var(y_{t+1}) + Var(\hat{y}_{t+1|t}) + E[y_{t+1}]^2 - 2E[y_{t+1}]E[\hat{y}_{t+1|t}] + E[\hat{y}_{t+1|t}]^2 \\ &= Var(y_{t+1}) + Var(\hat{y}_{t+1|t}) + \underbrace{(E[y_{t+1}] - E[\hat{y}_{t+1|t}])^2}_{Bias(\hat{y}_{t+1|t})^2} \\ &= Var(y_{t+1}) + Var(\hat{y}_{t+1|t}) + (Bias(\hat{y}_{t+1|t}))^2. \end{aligned}$$

Preliminary: Bias-Variance Tradeoff

- Consider two forecasts $\hat{y}_{t+1|t}$ and $\check{y}_{t+1|t}$ with the same MSFE:

$$\begin{aligned} \text{Var}(y_{t+1}) + \text{Var}(\hat{y}_{t+1|t}) + (\text{Bias}(\hat{y}_{t+1|t}))^2 &= \text{MSFE}(\hat{y}_{t+1|t}) \\ &= \text{MSFE}(\check{y}_{t+1|t}) = \text{Var}(y_{t+1}) + \text{Var}(\check{y}_{t+1|t}) + (\text{Bias}(\check{y}_{t+1|t}))^2 \\ \therefore \text{Var}(\hat{y}_{t+1|t}) + (\text{Bias}(\hat{y}_{t+1|t}))^2 &= \text{Var}(\check{y}_{t+1|t}) + (\text{Bias}(\check{y}_{t+1|t}))^2 \end{aligned}$$

- If $|\text{Bias}(\hat{y}_{t+1|t})| > |\text{Bias}(\check{y}_{t+1|t})|$, then $\text{Var}(\hat{y}_{t+1|t}) < \text{Var}(\check{y}_{t+1|t})$.
- If $|\text{Bias}(\hat{y}_{t+1|t})| < |\text{Bias}(\check{y}_{t+1|t})|$, then $\text{Var}(\hat{y}_{t+1|t}) > \text{Var}(\check{y}_{t+1|t})$.

Local Projections

Definition

Definition: The Object of Interest

- Consider a bivariate time-series data $\{(x_s, z_s)\}_{s=1}^t$.
- x_t and z_t are subject to “shocks” ε_t^x and ε_t^z , respectively.
- Denote $Y_t := [x_t \ z_t]'$, $\varepsilon_t := [\varepsilon_t^x \ \varepsilon_t^z]'$, and $\mathcal{I}_t := \{(x_s, z_s)\}_{s=1}^t$.
- The object of interest:

$$\begin{aligned} IRF_t(h) &:= \frac{\partial E[Y_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t} \\ &= \begin{bmatrix} \frac{\partial E[x_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^x} & \frac{\partial E[z_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^x} \\ \frac{\partial E[x_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^z} & \frac{\partial E[z_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^z} \end{bmatrix}. \end{aligned}$$

Definition: Structural Vector Autoregression Model

- We need a model to link (x_{t+h}, z_{t+h}) and $\mathcal{I}_t$.
 - intratemporal and intertemporal relationships
- Structural Vector Autoregression Model (SVAR):

$$x_t = \alpha_0 + \alpha_1 z_t + \alpha_2 x_{t-1} + \alpha_3 z_{t-1} + \varepsilon_t^x$$

$$z_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 z_{t-1} + \varepsilon_t^z,$$

where $\varepsilon_t^x \sim WN(0, \sigma_{\varepsilon^x}^2)$ and $\varepsilon_t^z \sim WN(0, \sigma_{\varepsilon^z}^2)$ with $Cov(\varepsilon_t^x, \varepsilon_t^z) = 0$.

Definition: Structural Vector Autoregression Model

- Structural Vector Autoregressive Model (SVAR):

$$A_0 Y_t = A_1 + A_2 Y_{t-1} + \varepsilon_t,$$

where

$$Y_t := \begin{bmatrix} x_t \\ z_t \end{bmatrix}, \quad A_0 := \begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}, \quad A_1 := \begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix}, \quad A_2 := \begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix},$$

and

$$\varepsilon_t := \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix} \quad \text{with} \quad \Xi_\varepsilon := \text{Var}(\varepsilon_t) = \begin{bmatrix} \sigma_{\varepsilon^x}^2 & 0 \\ 0 & \sigma_{\varepsilon^z}^2 \end{bmatrix}.$$

- A_0 , A_1 , A_2 , and Ξ_ε : structural parameters

Definition: Reduced-Form Vector Autoregression Model

- The reduced-form VAR model:

$$Y_t = B_0 + B_1 Y_{t-1} + \eta_t$$

where

$$B_0 := A_0^{-1}A_1, \quad B_1 := A_0^{-1}A_2, \quad \eta_t = \begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix} := A_0^{-1}\varepsilon_t,$$

and

$$\Xi_\eta := \text{Var}(\eta_t) = A_0^{-1}\Xi_\varepsilon(A_0^{-1})'.$$

- B_0 , B_1 , and Ξ_η : reduced-form parameters

Definition: The Object of Interest Redefined

- By repeated substitution,

$$Y_t = B_0 + B_1 Y_{t-1} + \eta_t$$
$$\implies Y_{t+h} = \left(\sum_{j=0}^{h-1} B_1^j \right) B_0 + B_1^{h+1} Y_{t-1} + B_1^h \underbrace{\eta_t}_{A_0^{-1} \epsilon_t} + \sum_{j=0}^{h-1} B_1^j \underbrace{\eta_{t+h-j}}_{A_0^{-1} \epsilon_{t+h-j}}.$$

$$- E[Y_{t+h} | \mathcal{I}_{t-1} \cup \{\epsilon_t\}] = \left(\sum_{j=0}^{h-1} B_1^j \right) B_0 + B_1^{h+1} Y_{t-1} + B_1^h A_0^{-1} \epsilon_t$$

$$- E[Y_{t+h} | \mathcal{I}_{t-1} \cup \{\epsilon_t + \delta\}] = \left(\sum_{j=0}^{h-1} B_1^j \right) B_0 + B_1^{h+1} Y_{t-1} + B_1^h A_0^{-1} (\epsilon_t + \delta)$$

Hence,

$$IRF_t(h) = B_1^h A_0^{-1}.$$

Vector Autoregressive (VAR) Models

Identification

Identification: Overview

- We want to recover $IRF_t(h) = B_1^h A_0^{-1}$.
- **Idea:** Directly recover B_1^h .
- Our approach proceeds in two steps:
 1. Reduced-form parameter B_1^h .
 2. Structural parameter A_0 .

Identification: Reduced-Form Parameters

- The reduced-form VAR

$$Y_t = B_0 + B_1 Y_{t-1} + \eta_t.$$

- By repeated substitution,

$$\begin{aligned} Y_{t+h} &= \underbrace{\left(\sum_{j=0}^{h-1} B_1^j \right) B_0}_{C_0} + \underbrace{B_1^h}_{C_1} Y_t + \underbrace{\sum_{j=0}^h B_1^j \eta_{t+h-j}}_{\nu_t} \\ &= C_0 + C_1 Y_t + \nu_t \\ &= \underbrace{\begin{bmatrix} C_0 & C_1 \end{bmatrix}}_{C'} \underbrace{\begin{bmatrix} 1 \\ Y_t \end{bmatrix}}_{\tilde{X}_t} + \nu_t \\ &= C' \tilde{X}_t + \nu_t. \end{aligned}$$

- This form is called the **direct regression** or **local projection** (Jordá, 2005).

Identification: Reduced-Form Parameters

Assumption 1 (TS1'): Linearity

A weakly stationary and weakly dependent stochastic process $\{(Y_t, \tilde{X}_t)\}_{t=0}^{T-h}$ follows

$$Y_{t+h} = C' \tilde{X}_t + \nu_t,$$

where $\{\nu_t\}$ is a sequence of reduced-form errors.

- By construction,

$$C = \begin{bmatrix} C_0 & C_1 \end{bmatrix}' = \begin{bmatrix} c_{0,1} & c_{1,1} & c_{1,2} \\ c_{0,2} & c_{1,3} & c_{1,4} \end{bmatrix}' = \begin{bmatrix} c_{0,1} & c_{0,2} \\ c_{1,1} & c_{1,3} \\ c_{1,2} & c_{1,4} \end{bmatrix}$$

Identification: Reduced-Form Parameters

Assumption 2 (TS2'): No Perfect Collinearity

None of the explanatory variables is a perfect linear combination of the others.

Assumption 3 (TS3'): Contemporaneous Exogeneity

$E[\nu_t | \tilde{X}_t] = \mathbf{0}$ for all t .

- $$E[\nu_t | \tilde{X}_t] = \sum_{j=0}^h B_1^j E[\eta_{t+h-j} | \tilde{X}_t] = \sum_{j=0}^h B_1^j A_0^{-1} E[\varepsilon_{t+h-j} | \tilde{X}_t].$$

Identification: Reduced-Form Parameters

Proposition: Identification of C

Under TS1'–TS3', the local-projection parameter matrix C is identified.

Proof. Taking transposes of TS1',

$$Y'_{t+h} = \tilde{X}'_t C + \nu'_t.$$

Premultiplying X_t and taking expectation,

$$E[\tilde{X}_t Y'_{t+h}] = E[\tilde{X}_t \tilde{X}'_t] C + \underbrace{E[\tilde{X}_t \nu'_t]}_{\mathbf{0} \text{ by TS3'}}.$$

In view of TS2', $(E[\tilde{X}_t \tilde{X}'_t])^{-1}$ exists, so that

$$C = (E[\tilde{X}_t \tilde{X}'_t])^{-1} E[\tilde{X}_t Y'_{t+h}].$$

Identification: Reduced-Form Parameters

Corollary: Identification

Under TS1'–TS3', the reduced-form parameter matrices B_0 and B_1 are identified.

Proof. By definition,

$$C_0 = B_0 \sum_{j=0}^{h-1} B_1^j \text{ and } C_1 = B_1^h.$$

B_1 and B_0 are identified as

$$B_1 = C_1^{\frac{1}{h}},$$

and

$$B_0 = C_0 \left(\sum_{j=0}^{h-1} B_1^j \right)^{-1}.$$



Identification: Reduced-Form Parameters

Corollary: Identification

Under TS1'–TS3', the reduced-form variance-covariance matrix Ξ_η is identified.

Proof. Plugging the identified C back into TS1',

$$\eta_t = Y_t - C' \tilde{X}_t,$$

so that

$$\Xi_\eta = E[(\eta_t - E[\eta_t])(\eta_t - E[\eta_t])']$$

is identified. □

Identification: Structural Parameters

- The reduced-form parameters B_0 , B_1 , and Ξ_η are identified.
- The structural parameters A_0 , A_1 , A_2 , and Ξ_ϵ can be identified by
 1. Normalization
 2. Short-run restrictions
 3. Long-run restrictions

Identification: Impulse Response Functions

- The object of interest is identified:

$$IRF_t(h) = B_1^h A_0^{-1} = C_1 A_0^{-1}.$$

- VAR method and LP method identify the same object of interest.
- $IRF_t(h)$ can be written in terms of A_0 and C_1 .

Local Projections

Estimation and Inference

Estimation and Inference: Reduced-Form Parameters

- **Idea:** We only need to estimate B_1^h (i.e., no need to estimate B_1 per se).
- Given the identification result

$$C = (E[\tilde{X}_t \tilde{X}_t'])^{-1} E[\tilde{X}_t Y_{t+h}'],$$

the sample-analog estimator is given by

$$\hat{C} := \left(\frac{1}{T-h} \sum_{t=0}^{T-h} \tilde{X}_t \tilde{X}_t' \right)^{-1} \left(\frac{1}{T-h} \sum_{t=0}^{T-h} \tilde{X}_t Y_{t+h}' \right).$$

- Since $C := B^h$, $\hat{C}$ directly estimates B^h .

Estimation and Inference: Reduced-Form Parameters

- LP method:

$$\begin{aligned}\hat{C} &= \left(\frac{1}{T-h} \sum_{t=1}^{T-h} X_t X_t' \right)^{-1} \left(\frac{1}{T-h} \sum_{t=1}^{T-h} X_t Y_{t+h}' \right) \\ &= C + \underbrace{\left(\frac{1}{T-h} \sum_{t=1}^{T-h} X_t X_t' \right)^{-1} \left(\frac{1}{T-h} \sum_{t=1}^{T-h} X_t \nu_{t+h}' \right)}_{=:\tau_C} = C + \tau_C.\end{aligned}$$

- VAR method:

$$\begin{aligned}\hat{B}^h &= \left\{ \left(\frac{1}{T} \sum_{t=1}^T X_t X_t' \right)^{-1} \left(\frac{1}{T} \sum_{t=1}^T X_t Y_t' \right) \right\}^h \\ &= \left\{ B + \underbrace{\left(\frac{1}{T} \sum_{t=1}^T X_t X_t' \right)^{-1} \left(\frac{1}{T} \sum_{t=1}^T X_t \eta_t' \right)}_{=:\tau_B} \right\}^h = (B + \tau_B)^h.\end{aligned}$$

Estimation and Inference: Structural Parameters

- The identification result gives

$$A_0 = g(C).$$

- The sample-analog estimator:

$$\hat{A}_0^{LP} = g(\hat{C}).$$

Estimation and Inference: Impulse Response Functions

- LP method:

$$\widehat{IRF}_t(h)^{LP} := \widehat{C}_1 (\widehat{A}_0^{LP})^{-1}.$$

- VAR method:

$$\widehat{IRF}_t(h)^{VAR} := \widehat{B}_1^h (\widehat{A}_0^{VAR})^{-1}.$$

- Q. Which estimator should be preferred?

Estimation and Inference: Impulse Response Functions

- Simulation study by Li et al. (2024):
 - In many practically-relevant setups,

$$|Bias(\widehat{IRF}_t(h)^{LP})| < |Bias(\widehat{IRF}_t(h)^{VAR})|.$$

- In view of the bias-variance tradeoff,

$$Var(\widehat{IRF}_t(h)^{LP}) > Var(\widehat{IRF}_t(h)^{VAR}).$$

- $\widehat{IRF}_t(h)^{LP}$ is more accurate, but less precise.
- $\widehat{IRF}_t(h)^{VAR}$ is less accurate, but more precise.

Estimation and Inference: Caveats

Large-sample properties:

- $\widehat{IRF}_t(h)^{LP}$ is consistent for $IRF_t(h)$ and asymptotically normally distributed.
 $\hookrightarrow \widehat{IRF}_t(h)^{LP}$ is as accurate as $\widehat{IRF}_t(h)^{VAR}$.

Variance estimator:

- Recall:

$$Y_{t+h} = C' \tilde{X}_t + \nu_t \quad \text{with} \quad \nu_t := \sum_{j=0}^h B_1^j \eta_{t+h-j} = \sum_{j=0}^h B_1^j A_0^{-1} \varepsilon_{t+h-j}$$

- $\{\nu_t\}$ have serial correlation and heteroskedasticity.
 $\hookrightarrow$ The Newey-West (or HAC) estimator!

Robustness to misspecification:

- LP is more robust to misspecification than VAR (Plagborg-Møller and Wolf, 2021).

Summary and Preview

Summary

- We studied the local projections (LP).
- LP replaces the iterated substitution with a single regression.
- In a finite sample, $\widehat{IRF}_t(h)^{LP}$ is more accurate, but less precise than $\widehat{IRF}_t(h)^{VAR}$.
- In a large sample, $\widehat{IRF}_t(h)^{LP}$ and $\widehat{IRF}_t(h)^{VAR}$ are consistent and normally distributed.
- For the variance estimation in LP, use the Newey-West estimator!

Preview

- Next week, we will review what we have studied this semester.

- Li, Dake, Mikkjel Plagborg-Møller, and Christian K. Wolf**, “Local projections vs. VARs: Lessons from thousands of DGPs,” *Journal of Econometrics*, 2024, 244 (2), 105722.
- Óscar Jordá**, “Estimation and Inference of Impulse Responses by Local Projections,” *American Economic Review*, 2005, 95 (1), 161–182.
- Plagborg-Møller, Mikkjel and Christian K. Wolf**, “Local Projections and VARs Estimate the Same Impulse Responses,” *Econometrica*, 2021, 89 (2), 955–980.